

BT / OPEN RESEARCH PROJECT

Microfluidic Screening for Antimicrobial Resistance

Design a low-volume microfluidic assay concept for rapidly comparing bacterial response to antibiotic concentration gradients under supervised laboratory conditions.

TEAM	DURATION	FORMAT
Up to 3 students	14-week prototype	Mentored research

Problem

Conventional susceptibility workflows can be resource intensive. A carefully validated microfluidic approach could reduce reagent use and shorten experimental turnaround while preserving biosafety and interpretability.

Method

The project begins with non-pathogenic teaching strains and simulated image data. Students will design channel geometry, model diffusion, develop an image-analysis pipeline and prepare a biosafety-reviewed protocol before any wet-lab work.

Research objectives

- Simulate stable antibiotic concentration gradients
- Develop transparent growth/no-growth image measurements
- Define positive, negative and contamination controls
- Complete biosafety and ethical review before wet-lab validation

Skills and preparation

- Biotechnology laboratory fundamentals
- CAD or microfluidic design
- Microscopy image analysis
- Biosafety documentation

Expected deliverables

- Microfluidic design files
- Diffusion simulation report
- Image-analysis prototype
- Biosafety-reviewed validation protocol

Indicative timeline

Weeks 1-3	review and biosafety planning
Weeks 4-6	device simulation
Weeks 7-10	image-analysis prototype
Weeks 11-14	supervised validation plan

Responsible-use note

This brief describes a staging research concept. It does not promise funding, credit, authorship, laboratory access, safety clearance or selection. The faculty lead must approve the final protocol, data access, ethics and safety requirements before work begins.

Apply through the authenticated NSUT Connect project page. Do not send passwords, secret keys, medical information or confidential research data with an expression of interest.